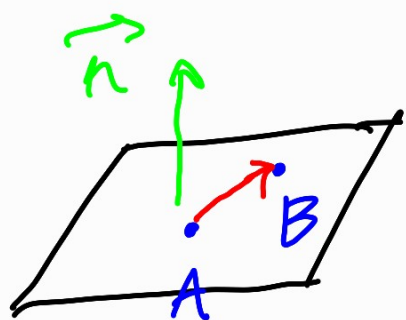


If you are new to the class,
please check announcement on 9/1
on lecture site on Canvas
Classes with document camera:
will switch to direct feed next
week once new tablet arrives
Feedback survey in 2±1 weeks

8: Plane through $\overset{A \rightarrow}{(0, -2, 5)}$ & $\overset{B \rightarrow}{(-1, 3, 1)}$
perpendicular to $5x + 4y - 2z = 0$

Recall that it suffices to find
a point on the plane & a normal
vector. We already have a point.



Let $\vec{n}$ be a normal
vector to the plane.

Note that $\vec{n}$ is perpendicular to
any vector in the plane, so

$$\vec{n} \perp \vec{AB} = A - B = \langle 1, -5, 4 \rangle.$$

Also note perpendicular planes have perpendicular normal vectors, so

$$\vec{n} \perp \langle 5, 4, -2 \rangle \text{ since this is the normal to } 5x + 4y - 2z = 0.$$

$$\text{Thus, } \langle 1, -5, 4 \rangle \times \langle 5, 4, -2 \rangle = \langle -6, 22, 29 \rangle$$

$$\begin{array}{l} \left| \begin{array}{ccc} 1 & -5 & 4 \\ 5 & 4 & -2 \end{array} \right| \quad \begin{array}{l} i: (-5)(-2) - 4 \cdot 4 = 10 - 16 = -6 \\ j: -[1(-2) - 5 \cdot 4] = -(-2 - 20) = 22 \\ k: 1 \cdot 4 - (-5) \cdot 5 = 4 + 25 = 29 \end{array} \end{array}$$

is a normal vector to the plane.

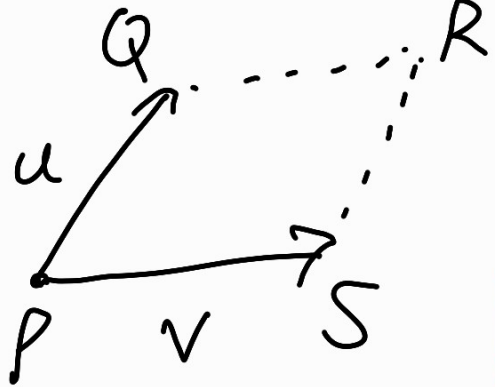
Since $\langle -6, 22, 29 \rangle$ is a normal & the plane contains $(0, -2, 5)$,

$$\boxed{-6x + 22(y + 2) + 29(z - 5) = 0}$$

is one possible plane equation.

$$3: P = (1, 2, 1), Q = (2, 5, 4), R = (6, 9, 12), S = (5, 6, 9)$$

(2) We can show P, Q, R, S form a parallelogram and hence they're coplanar.



Let $u = \overrightarrow{PQ}$, $v = \overrightarrow{PS}$.

Then $u = Q - P = \langle 1, 3, 3 \rangle$,
 $v = S - P = \langle 4, 4, 8 \rangle$.

A quadrilateral is a parallelogram iff 2 opposite sides are represented by the same vector, so we just need to check that $u = \overrightarrow{SR}$ or $v = \overrightarrow{QR}$.

Indeed, $\overrightarrow{SR} = R - S = \langle 1, 3, 3 \rangle = u$.

(6) Recall that $\|u \times v\|$ represents the area of a parallelogram spanned by u & v ,

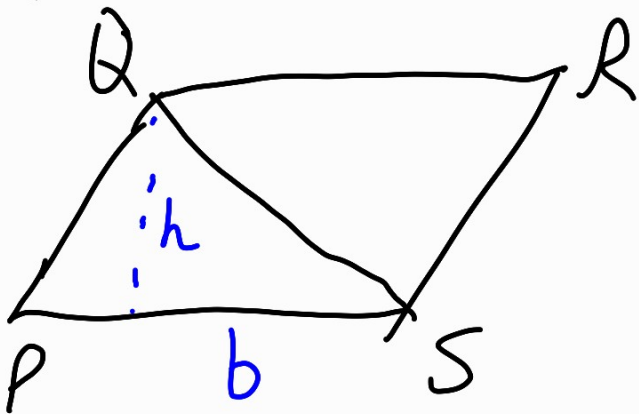
so $[PQRS] = \|u \times v\|$.

Notice $u \times v = 4 (\langle 1, 3, 3 \rangle \times \langle 4, 4, 8 \rangle)$
 $= 4 \langle 3, 1, -2 \rangle$
 $= \langle 12, 4, -8 \rangle$

$$\begin{array}{l} 133 \quad \therefore 3 \cdot 2 - 3 \cdot 1 = 3 \\ 112 \quad \therefore -(1 \cdot 2 - 1 \cdot 3) = 1 \\ u: \quad 1 \cdot 1 - 1 \cdot 3 = -2 \end{array}$$

$$\|u \times v\| = 4|\langle 3, 1, -2 \rangle| = 4\sqrt{14}.$$

(c) Notice $[PQS] = \frac{1}{2}[PQRS] = 2\sqrt{14}$ because PQS & PQRS have the same base & height but there is a $\frac{1}{2}$ factor in the area of a triangle formula. Alternatively, PQRS is composed of 2 copies of PQS.



1d: $\langle 1, -1, 3 \rangle \times \langle 2, -2, 6 \rangle = 0$
 $\langle 1, 2, 3 \rangle \cdot \langle -3, 0, 1 \rangle = 0$

This is suggesting that parallel vectors have cross product 0, perpendicular vectors have dot product 0, and the dot for b, x for c similarly suggest properties of $\cdot$ for parallel

and $\times$ for perpendicular vectors.

$$2: \quad \mathbf{a} = \langle 2, 1, -3 \rangle, \quad \mathbf{b} = \langle 0, -2, 1 \rangle$$

$$(2) \quad \begin{array}{l} 2, 1, -3 \\ 0, -2, 1 \end{array} \quad \begin{array}{l} i: 1 \cdot 1 - (-2)(-3) = 1 - 6 = -5 \\ j: -(2 \cdot 1 - 0(-3)) = -2 \\ k: 2(-2) - 0 \cdot 1 = -4 \end{array}$$

$$\text{so} \quad \mathbf{a} \times \mathbf{b} = \langle -5, -2, -4 \rangle$$

(b) need to check that dot product is 0 to verify orthogonality

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = \langle -5, -2, -4 \rangle \cdot \langle 2, 1, -3 \rangle = -10 - 2 + 12 = 0$$

$$(c) \quad \text{First check } |\langle -5, -2, -4 \rangle| = \sqrt{45} = 3\sqrt{5}. \quad \text{Now notice}$$

$$\mathbf{u} = \frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} = \frac{\langle -5, -2, -4 \rangle}{3\sqrt{5}} = -\frac{1}{3\sqrt{5}} \langle 5, 2, 4 \rangle$$

is also orthogonal to $\mathbf{a}$ & $\mathbf{b}$ since scaling doesn't affect orthogonality.

So $\mathbf{u}$ & $-\mathbf{u}$ are 2 such unit vectors.

4: Line through $P = (2, 2, 4)$ parallel to $\langle 4, -1, 7 \rangle$.

(2) $\langle 4, -1, 7 \rangle t = \langle 4t, -t, 7t \rangle$ represents a line, and is always parallel to $\langle 4, -1, 7 \rangle$.

You can now shift this line to pass through $(2, 2, 4)$ by adding on $\langle 2, 2, 4 \rangle$ to get

$$\langle 4t + 2, -t + 2, 7t + 4 \rangle$$

(6) From the vector equation,

$$x = 4t + 2, y = -t + 2, z = 7t + 4$$

are parametric equations.

(c) To get symmetric equations, solve for t in terms of x , y , and z . So

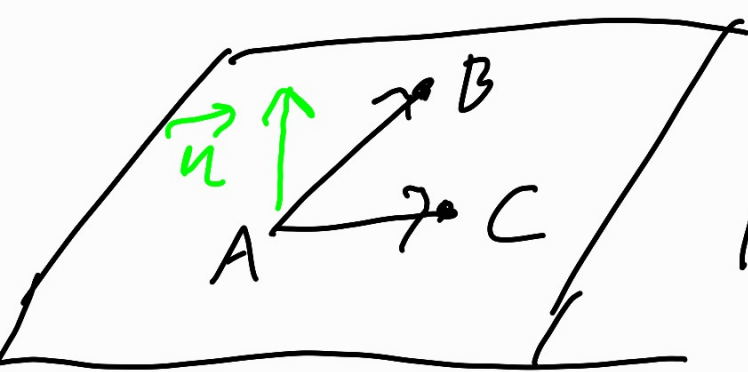
$$x = 4t + 2 \Rightarrow t = \frac{x-2}{4}$$

$$= \frac{y-2}{-1} = \frac{z-4}{7},$$

and all of a sudden the symmetric equations appear.

$\mathcal{L}$: Plane through $A = (0, 1, 2)$,
 $B = (1, 2, 3)$, $C = (2, 3, 5)$

To find $\perp$ plane, need point + normal vector. Let $\vec{n}$ be a normal vector to the plane.



$\vec{n}$ will be perpendicular to

any vector inside the plane,
so in particular $\vec{n} \perp \vec{AB}, \vec{AC}$.

Thus, we can choose

$$\vec{n} = \vec{AB} \times \vec{AC} := u \times v$$

$$u = B - A = \langle 1, 1, 1 \rangle$$

$$v = C - A = \langle 2, 2, 3 \rangle.$$

Now notice that

$$v = \langle 2, 2, 2 \rangle + \langle 0, 0, 1 \rangle = 2u + w,$$

$$\text{so } u \times v = u \times (2u + w)$$

$$= 2(u \times u) + u \times w$$

$$= u \times w$$

$$= \langle 1, -1, 0 \rangle$$

$$\begin{array}{ccc|ccc} 1 & 1 & 1 & \vdots & \vdots & \vdots \\ 0 & 0 & 1 & \vdots & \vdots & \vdots \\ & & & \vdots & \vdots & \vdots \end{array} \begin{array}{l} 1 \\ -1 \\ 0 \end{array}$$

So we know the plane has normal $\langle 1, -1, 0 \rangle$ & passes through $(0, 1, 2)$, so its equation is

$$0 = 1(x-0) - (y-1) + 0(z-2)$$

$$= x - y + 1 \Rightarrow x = y - 1.$$

6: Determine parallel, intersecting, or skew

(2) Recall that 2 lines are parallel iff their direction vectors are parallel.

$L_1: r(t) = \langle -1, 2, 3 \rangle + \langle 3, 4, -2 \rangle t$
is parallel to $\langle 3, 4, -2 \rangle t$,
which is parallel to $\langle 3, 4, -2 \rangle$.

L_2 : First convert the line to vector form it, and then we can extract the direction.

$$\text{Set } t = \frac{x-1}{2} = \frac{y}{-3} = \frac{z+1}{-3},$$

$$\text{so } x = 2t + 1, y = -3t, z = -3t + 1$$

$$\Rightarrow \langle 2t+1, -3t, -3t+1 \rangle,$$

which is parallel to $\langle 2, -3, -3 \rangle$,
represents L_2 .

$\langle 3, 4, -2 \rangle$ & $\langle 2, -3, -3 \rangle$ are
not parallel, so L_1 & L_2 are
not parallel.

To check if 2 lines intersect,
plug the equation for one line
into the other & see if there
are any solutions.

L_1 has $x = -1 + 3t$, $y = 2 + 4t$,
 $z = 3 - 2t$, plug this into L_2
to get

$$\frac{-2+3t}{2} = \frac{x-1}{2} = \frac{y}{-3} = \frac{2+4t}{-3}$$

$$\Rightarrow -3(-2+3t) = 2(2+4t),$$

$$\text{so } 6-9t = 4+8t \Rightarrow 2=17t$$

$$\Rightarrow t = 2/17$$

$$\frac{2+4t}{-3} = \frac{y}{-3} = \frac{z+1}{-3} = \frac{4-2t}{-3}$$

$$\Rightarrow 2+4t = 4-2t \Rightarrow 6t=2$$

$$\Rightarrow t = 1/3$$

But $\frac{2}{17} \neq \frac{1}{3}$, so it's impossible for both equations for L_2 to be true, so L_1 & L_2 can't intersect, so skew.

(6) L_1 is parallel to $\langle 2, 1, -1 \rangle$
from looking at the t
coefficients, L_2 is parallel
to $\langle -3, -\frac{3}{2}, \frac{3}{2} \rangle$ from
looking at the s coefficients.

$$\langle -3, -1.5, 1.5 \rangle = -1.5 \langle 2, 1, -1 \rangle,$$

so these 2 vectors are

parallel, so L_1 & L_2 are parallel.
